Answers

Worksheet

Checking and securing understanding of non-linear sequences

Task A: Arithmetic and geometric sequences

1) Categorise these sequences into arithmetic and geometric.

Arithmetic

a) $-4, -2, 0, 2, \ldots$

Common difference $+2$

e) $36, 24, 12, 0, \ldots$

Common difference -12

c) $-4, 4, 12, 20, \ldots$

Common difference $+8$

h) $1, -2, -5, -8, \ldots$

Common difference -3

Geometric

b) $2, 4, 8, 16, \ldots$

Common ratio 2

f) $108, 36, 12, 4, \ldots$

Common ratio $\frac{1}{3}$

d) $4, 12, 36, 108, \ldots$

Common ratio 3

g) $1, -2, 4, -8, \ldots$

Common ratio -2

2) Find the blanked out 1^{st} and 4^{th} terms of these sequences.

a) The arithmetic sequence $-9,\ 9,\ 27,\ 45, \ldots$

$+18 \quad +18 \quad +18$

b) The geometric sequence $3,\ 9,\ 27,\ 81, \ldots$

$\times 3 \quad \times 3 \quad \times 3$

c) The arithmetic sequence $7,\ 4,\ 1,\ -2, \ldots$

$-3 \quad -3 \quad -3$

d) The geometric sequence $16,\ 4,\ 1,\ \frac{1}{4}, \ldots$

$\times\frac{1}{4} \quad \times\frac{1}{4} \quad \times\frac{1}{4}$

Task B: Other types of sequences

1) Identify which of these sequences have a common second difference, and which have a common third difference.

a) $1, 2, 12, 31, 59, \dots$

$+1 \quad +10 \quad +19 \quad +28$

$+9 \quad +9 \quad +9$

Common second difference $+9$

b) $1, 2, 5, 11, 21, \dots$

$+1 \quad +3 \quad +6 \quad +10$

$+2 \quad +3 \quad +4$

$+1 \quad +1$

Common third difference $+1$

c) $100, 95, 86, 65, 24, \dots$

$-5 \quad -9 \quad -21 \quad -41$

$-4 \quad -12 \quad -20$

$-8 \quad -8$

Common third difference -8

d) $-68, -66, -54, -32, 0, \dots$

$+2 \quad +12 \quad +22 \quad +32$

$+10 \quad +10 \quad +10$

Common second difference $+10$

2) Given that these sequences have a common second difference find the missing terms.

a) $1, 5, 11, 19, 29, \dots$

$+4 \quad +6 \quad +8 \quad +10$

$+2 \quad +2 \quad +2$

b) $12, 14, 23, 39, 62, \dots$

$+2 \quad +9 \quad +16 \quad +23$

$+7 \quad +7 \quad +7$

c) $0, 20, 36, 48, 56, \dots$

$+20 \quad +16 \quad +12 \quad +8$

$-4 \quad -4 \quad -4$

d) $\frac{1}{4}, \frac{1}{2}, \frac{5}{4}, \frac{5}{2}, \frac{17}{4}, \dots$

$+\frac{1}{4} \quad +\frac{3}{4} \quad +\frac{5}{4} \quad +\frac{7}{4}$

$+\frac{1}{2} \quad +\frac{1}{2} \quad +\frac{1}{2}$

3) Jacob is examining these three sequences.

a) 12, 20, 28, 36, 44, …

b) 12, 20, 30, 42, 56, …

c) 56, 70, 82, 92, 100, …

I see three increasing sequences whereby the terms will always go up.

Jacob's statement is not entirely true. Explore the sequences and see if you can find the contradiction.

a) 12, 20, 28, 36, 44, 52, 60, 68, 76, 84, …

+8 +8 +8 +8 +8 +8 +8 +8 +8

Arithmetic sequence, positive common difference. Terms will increase forever.

b) 12, 20, 30, 42, 56, 82, 100, 120, 142, …

+8 +10 +12 +1 +16 +18 +20 +22

+2 +2 +2 +2 +2 +2 +2

Positive second difference forever increasing the first difference. Terms will increase forever.

c) 56, 70, 82, 92, 100, 106, 110, 112, 112, 110, 106, 100, 92, …

+14 +12 +10 +8 +6 +4 +2 +0 −2 −4 −6 −8

−2 −2 −2 −2 −2 −2 −2 −2 −2 −2 −2

Negative second difference decreasing the first difference.